

(Writing Sample)

Does democracy bring economic growth?

Evidence from Selected Developing Countries

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Abstract:

This research examines whether democracy brings economic growth to developing countries. Using the democracy index as a proxy variable for democratic quality, this paper discusses its effect on economic growth. After adding lagged values of economic growth into the models to control for economic dynamics, this research finds that higher democracy scores are associated with better economic performance. However, concerns about omitted variable bias and heterogeneity still exist in the model. Considering broader control variables, suitable instrumental variables, and more complete data in future research may help us better understand the effect of democracy on economic growth. The results of this research may help countries and international organizations understand the relationship between democracy and economic development. They may also provide a direction for future policy research on whether democratic institutions should be promoted together with other development policies, such as education, investment, and political stability.

Section 1: Introduction

After World War II many countries decolonized and became independent states. Over the past 60 to 70 years, many international organizations such as United Nations, World Bank, and various NGOs have invested lots of money and resources in these countries tried to reduce poverty and promote economic development. Many studies suggest that inclusive institutions, such as democracy, can bring economic growth. Moreover, most developed countries in the world today are democratic countries, which has encouraged international organizations to learn from their experience and promote democracy institution in these developing and undeveloped countries. However, many of these countries, especially in Southeast Asia and Sub-Saharan Africa remain at relatively low level of economic development today.

This paper examines whether higher levels of democracy are associated with better economic performance among selected developing countries. Using an existing democracy index dataset, this research analyzes the impact of democracy on real GDP per capita growth and examine whether the democracy index has statistically significant association with economic performance. If the result shows that democracy does not have a statistically significant effect on growth rate, this may suggest that the relationship between democracy and economic growth may be more context-dependent than a simple institutional-growth argument implies. Such findings could imply international organizations should allocate their resources to different aspect of development rather than focusing solely on promoting democratic institutions.

The results of this paper suggest that democracy is positively associated with economic performance. After controlling for lagged values of real GDP per capita growth, country fixed effects, year fixed effects, and other economic controls, higher democracy scores are associated with higher real GDP per capita growth. However, the interaction between democracy and education is not statistically significant, and the lagged-democracy robustness checks do not provide strong evidence of delayed effects. Therefore, the findings support a positive association between democracy and economic growth, but they should not be interpreted as definitive causal evidence.

Section 2: Literature Review

Acemoglu et al. (2019) examine whether democratization has a causal effect on long-run economic growth. While some argue that autocracy regimes may promote faster economic expansion due to centralized decision-making, the authors test whether democracy leads to long-run economics growth over time.

They construct a binary democracy index by combining several democracy datasets and examine country-level data from 1960 to 2010. Because democratization often occurs during periods of worse economic condition, they include lagged GDP in their regressions to

account for pre-existing economic conditions and try to isolate the independent effect of democracy on subsequent economic rebound. To deal with potential endogeneity concerns, such as the possibility that richer countries are more likely to remain democratic, they employ multiple econometric strategies. They use regional waves of democratization as an instrument variable in a two-stage least squares framework.

Their main finding is that democratization increases GDP per capita by approximately 20–25 percent in the long run, measured over about 25 years after the democratic transition. The authors also test different mechanisms and find that democracy brings higher tax revenues as a share of GDP, lower child mortality, and higher school enrollment rates. These results suggest that democracy improves both economic performance and broader social outcomes.

On the other hand, the literature on authoritarian institutions and party organization shows that authoritarian regimes are not all the same. Blattman, Gehlbach, and Yu (2025) argue that the main economic problem of authoritarianism comes from personalist dictatorship, rather than authoritarian rule in general. Similarly, Gehlbach and Keefer (2011) suggest that even under authoritarian rule, high levels of private investment may still exist when ruling parties are highly institutionalized. These studies imply that not all authoritarian regimes weaken economic performance.

Section 3: Data

The GDP per capita data are sourced from the World Bank and report real GDP per capita (in constant 2015 U.S. dollars) for different countries from 1960 to 2024. I take the natural logarithm of real GDP per capita to discuss the effect of democracy on economic growth. This paper also constructs the annual growth rate of real GDP per capita as the first difference of log real GDP per capita multiplied by 100. The panel unit root test fails to reject the null hypothesis that log real GDP per capita contains a unit root, suggesting it's non-stationary in levels. However, the first difference of log real GDP per capita, which represents real GDP per capita growth rate, appears to be stationary.

The democracy index is obtained from Our World in Data, which compiles data from Polity V. The index ranges from -10 to 10 , where lower values indicate less democratic regimes and higher values indicate more democratic regimes. And this country level dataset covers the period from 1950 to 2018. In this research, I standardize democracy index to use in my regression called stdDem.

Trade openness and Gross fixed capital formation (GFCF) come from World Bank. Both are country level data and are available from 1960 to 2024. Trade openness is defined as the sum of exports and imports of goods and services, divided by GDP, and expressed as a percentage of GDP. GFCF measures investment in fixed productive assets, such as buildings,

infrastructure, and machinery, and is expressed as a share of GDP.

Finally, the education data are obtained from Our World in Data, which compiles data from Barro-Lee dataset. It reports the average years of schooling for each country from 1950 to 2020. This dataset is available at five-year interval. To construct a more complete dataset, I linearly interpolate missing education values within each country because relevant education policies often have delayed effects. For each country, missing values between two observed years are imputed using the linear change implied by the adjacent observed education values. Observed values are not altered.

Although the motivation focuses on developing countries in Sub-Saharan Africa and Southeast Asia, the final dataset consists of selected developing countries that remain after data cleaning. Each dataset covers different periods, so I restrict observations from 1960 to 2015, then drop observations with missing values in main variables of this research, that is Democracy and Real GDP per capita. I also drop six countries with extensive missing values in the control variables, including Somalia, Nigeria, Liberia, Burundi, Myanmar, and Malawi. Furthermore, Cambodia's real GDP per capita series has a break between 1978 and 1989. Because the regression includes lagged GDP, I drop Cambodia's observations before 1989 to avoid problems caused by this break in the time series. To deal with the missing-value problem in GFCF, trade openness, and education years, I set missing values equal to zero and construct a binary indicator for each variable. Each indicator equals one if the original value of the corresponding variable is missing and zero otherwise. I include these indicators in all regressions that use the corresponding control variables. This method keeps observations with missing control variables in the regression sample and controls for possible differences between observations with and without missing data.

The final dataset is a country-year panel with 2,492 observations. Because real GDP per capita growth rates and lagged growth variables require differencing and lagging construction, the regression sample will be smaller than the final dataset. The unit of observation is the country-year. Table 1 is table of variable definitions. Table 2 is summary statistics table about variable in models. Table 3 is about main variables' (growth rate and stdDem) detail information. Figures 1 and 2 illustrate the unconditional relationship between stdDem and growth rate. Figure 1 shows a binned scatter plot in which the stdDem is divided into fifteen bins. The fitted line is upward sloping, suggesting that higher democracy scores are associated with higher average growth rate. However, the slope is relatively modest, indicating that the unconditional linear relationship is positive but not very strong. Figure 2 presents a possible nonlinear relationship. It suggests that the relationship between democracy and growth rate is uneven across the range of democracy scores. The overall pattern still suggests that highly democratic observations tend to be associated with better economic performance. The possible nonlinearity relationship in Figure 2 suggests that the

baseline linear relationship should be interpreted as an average conditional association rather than a complete description of the democracy-economic growth relationship. Table 4 and table 5 show unit root test outcome of log real GDP per capita and growth rate, respectively.

Section 4: Model

$$y_{\{i,t\}} = \beta_0 + \beta_1 stdDem_{\{i,t\}} + \sum_{j=1} \gamma_j y_{\{i,t-j\}} + \beta_2 EDUC_{\{i,t\}} + \beta_3 GFCF_{\{i,t\}} + \beta_4 TradeOpen_{\{i,t\}} \\ + \beta_5 EDUCmiss_{\{i,t\}} + \beta_6 GFCFmiss_{\{i,t\}} + \beta_7 TradeOpenmiss_{\{i,t\}} + a_i + b_t \\ + \varepsilon_{\{i,t\}}$$

The dependent variable ($y_{i,t}$) in this paper is country i 's real GDP per capita growth rate in year t . The main regression includes below independent variables. First, $stdDem_{i,t}$ represents country i 's democracy index in year t which is the main variable of interest in this paper. $y_{i,t-j}$ means growth rate of country i , measured j years prior to year t . The reason for including $y_{i,t-j}$ (growth rate lag) is that Acemoglu et al. (2019) argue that GDP dynamics are important because democratization may occur after periods of economic decline. Therefore, including lagged dependent variables helps us distinguish whether subsequent economy recovery was caused by democratization or by natural business cycle fluctuation. Education is another variable which may affect GDP per capita. $EDUC_{i,t}$ represents country i 's average years of schooling in year t . $GFCF_{i,t}$ represents that country i 's gross fixed capital formation in year t which be interpreted as investment ratio or capital accumulation rate in this model. Barro (1991) indicated that investment ratio, capital accumulation, and education have positive effect to economic growth. $Tradeopen_{i,t}$ represents country i 's trade openness degree in year t . Yanikkata (2003) suggests that open trading could bring positive influence on economic growth. $EDUCmiss_{i,t}$, $GFCFmiss_{i,t}$, and $TradeOpenmiss_{i,t}$ are missing-value indicators. a_i is country i 's fixed effect. b_t is year t 's fixed effect. $\varepsilon_{i,t}$ is error term of my regression. The parameter of interest is β_1 , the coefficient on $stdDem$. I expect it to be positive since democratic institution may be associated with better economic performance through channels such as political accountability and more stable environment for investment.

$$y_{\{i,t\}} = \beta_0 + \beta_1 stdDem_{\{i,t\}} + \sum_{j=1} \gamma_j y_{\{i,t-j\}} + \beta_2 EDUC_{\{i,t\}} + \beta_3 GFCF_{\{i,t\}} + \beta_4 TradeOpen_{\{i,t\}} \\ + \beta_5 (EDUC_{\{i,t\}} * stdDem_{\{i,t\}}) + \beta_6 EDUCmiss_{\{i,t\}} + \beta_7 GFCFmiss_{\{i,t\}} \\ + \beta_8 TradeOpenmiss_{\{i,t\}} + a_i + b_t + \varepsilon_{\{i,t\}}$$

In the second model, I add one interaction term between democracy index ($stdDem$) and avg schooling year ($EDUC$). This interaction term discusses whether the relationship between democracy and economic performance varies with education years. Equivalently, it also

examines whether the relationship between years of schooling and economic performance varies with the level of democracy, holding other variables constant. In the former case, holding the level of democracy constant, the country with higher education levels, people may benefit more from democratic regime. More educated citizens may better understand the importance of participating in elections and monitoring government behavior. As a result, the accountability mechanism of democracy may work more effectively when education levels are higher. For the latter case, holding education levels constant, the country with more democratic institution than other countries may magnify positive effect of education. Democratic institutions may provide stronger protection of property rights, greater policy accountability, stabler environment, and more opportunities for economic participation. Therefore, the same level of education may generate larger economic benefits in a more democratic institutional country. From above reasons, I expect the coefficient on the interaction term between `stdDem` and `EDUC` to be positive.

This model may face several identification concerns, including reverse causality, omitted variable bias, and broader endogeneity problems. First, reverse causality may arise because economic performance can affect political institutions. For example, countries with low initial levels of development may experience faster growth during the development process, and sustained economic improvement may later contribute to political stability and democratization, especially we focus on Sub-African countries and Southeast Asia countries.

Second concern is about omitted variable bias. Factors such as civil conflict, civil war, regional or global democratization waves, natural disasters, global economic shocks, and other systematical changes may affect economic growth and the quality of democracy at the same time. If these factors are not being totally considered, the estimated coefficient on democracy may capture part of their influence rather than the effect of democracy itself.

To reduce these concerns, models include country fixed effects and year fixed effects. Country fixed effects control for time-invariant differences across countries. Year fixed effects control for shocks that affect all countries each year, such as financial crisis. I also include lagged growth rates to control for previous economic performance and growth dynamics. This helps reduce the possibility that the estimated effect of democracy simply reflects short-run economic recovery after earlier periods of poor performance. However, these controls do not fully eliminate endogeneity concerns, especially those caused by time-varying omitted factors or reverse causality. Therefore, the results should be interpreted as conditional associations rather than definitive causal estimates. In robustness check section, I will examine whether lagged democracy influences economic growth. This helps test whether democratic institutions or regime changes may influence economic performance with a delay rather than only through contemporaneous changes.

Section 5: Empirical Analysis

In model 1, I examine how many real GDP per capita growth rate lags we need to put into the model as my first step. From table 6, the specification with three lags appears appropriate. In the model with one to three lags, the third lag of real GDP per capita growth is statistically significant at the 1% level. Additional lags beyond the third lag are not statistically significant, so I use three growth lags in the subsequent analysis. I estimate model 1 using three model specifications, without fixed effects, with country fixed effects, and with country and year fixed effects. From table 7, the result shows that coefficients of stdDem (.344, .438, and .413) are all positive and statistically significant under 1% significance level. The results are consistent with my prediction that higher democracy scores are associated with better economic performance. In the regression with time fixed effect and country fixed effect, the coefficient of stdDem equals to .413, which means one standard deviation increase in the democracy index is associated with a .413 percentage-point increase in real GDP per capita growth, holding other variables constant. Then, to evaluate economic significance, I calculate the standardized coefficient. The standardized effect is about .067 standard deviation of growth, which suggests that one standard deviation increase in democracy is associated with a relatively modest change in growth. Now, considering 25th and 75th percentile of stdDem which are about -.968 and .973, respectively. This interquartile-range change implies a standardized effect of about .13, which means that the interquartile-range effect is economically significant. This difference arises because the interquartile-range calculation evaluates a larger change in the democracy index than the one-standard-deviation calculation. As a result, the predicted effect on growth is larger. Therefore, the result suggests that small changes in democracy may have limited economic significance, while larger changes across the distribution of democracy may be economically meaningful.

As in model 1, before analyzing model 2, I first examine how many real GDP per capita growth rate lags should be included in model 2. From table 8, the result shows that with three lags is appropriate. In model 2, after putting three growth rate lags, I also consider three model specifications, without fixed effects, with country fixed effect, and with country and year fixed effect. From table 9, coefficients of stdDem are all positive (.263, .297, and .319) in these three specifications. Also, coefficients of interaction term between stdDem and EDUC are all positive (.0326, .0592, and .0334). The results are consistent with my previous prediction that interaction between democracy and education is associated with better economic performance. However, neither stdDem nor interaction term is statistically significant in model 2. The possible reason may come from the interpretation of stdDem changes after adding the interaction term. In Model 2, the coefficient on stdDem represents the effect of democracy when EDUC equals zero, rather than the overall average effect of democracy. Since $EDUC = 0$ is not substantively meaningful in this context, the coefficient

on stdDem alone should be interpreted cautiously. The marginal effect of democracy depends on education and is equal to coefficient of interaction term times EDUC plus coefficient of stdDem. Another possible explanation is multicollinearity among stdDem, EDUC, and their interaction term. Because the interaction term is composed from the two original variables, it may increase the standard errors of the estimates and reduce statistical significance. Thus, the insignificant result in Model 2 may reflect lower estimation precision. To evaluate the economic significance of model 2, I calculate the marginal effect of democracy at the average level of education. In the interaction model, the marginal effect of stdDem is $\beta_1 + \beta_5 \text{EDUC}$. Evaluated at the mean of EDUC, the standardized marginal effect is about 0.065, which means the effect of democracy at the average education level is not economically significant.

Table 10 reports Model 1 and Model 2 in the same table for easy reference. Column 1 presents the baseline model without the interaction term. Column 2 presents the interaction model, which adds $\text{stdDem} \times \text{EDUC}$. Both models include three lags of real GDP per capita growth, missing-value indicators, country fixed effects, and year fixed effects.

Section 6: Robustness

Because the effect of democratic institutions on economic performance may operate with a delay, I conduct a robustness check by adding lagged values of stdDem to model 1. Democratic institutions may affect growth through policy changes, investment decisions, and institutional adjustments, all of which may take time to influence economic outcomes. Table 11 reports several specifications using current and lagged values of stdDem to examine whether democracy has delayed effects on real GDP per capita growth. The result shows that lagged values of stdDem are not statistically significant for real GDP per capita growth. The possible reason is democracy scores may not change significantly from year to year unless there is a major shock, such as civil war or global democratization wave influence. As a result, current democracy scores may be highly correlated with previous periods.

Section 7: Conclusion

This research examines whether democracy has positive effect on a country's economic growth, especially this paper is motivated by developing countries in Sub-Saharan Africa and Southeast Asia. After controlling for year fixed effects, country fixed effects, and including three lags of real GDP per capita growth into models, the estimated effect of democracy is positive as I predict. However, two models in this research show different outcomes. The model without interaction term, democracy has a statistically significant positive effect at the 1% significant level. After adding the interaction term between democracy scores and education levels, the effect of democracy is no longer statistically significant for economic growth. In terms of economic significance, the one-standard deviation effect of democracy is

not economically significant. However, the interquartile-range effect in Model 1 suggests that larger changes in democracy score may be economically significant. In the robustness section, I examine whether democracy has a delayed effect on economic performance. Since institutional structures do not change frequently, democracy scores are highly persistent over time and are strongly correlated with values in previous periods. This persistence may make it difficult to separately estimate the effects of current and lagged scores.

Because this paper uses observational panel data, the results should be interpreted as conditional associations rather than definitive causality. Future research could use stronger identification strategies, such as suitable instrumental variables, democratization shocks, or regime transition event-study designs, and more complete data, to better identify the causal effect of democracy on growth.

The results suggest that higher democracy score may be associated with better economic performance, but democracy alone may not be sufficient to generate sustained economic growth. Future policies may need to combine democratic institution-building with investment in education, political stability, and private property protection.

References

- Acemoglu et al. 2019. "Democracy Does Cause Growth." *Journal of Political Economy* 127 (1): 47–100.
- Barro, Robert J. 1991. "Economic Growth in a Cross Section of Countries." *Quarterly Journal of Economics* 106(2): 407-433.
- Blattman, Christopher, Scott Gehlbach, and Zeyang Yu. 2025. "The Personalist Penalty: Varieties of Autocracy and Economic Growth." NBER Working Paper 34093. Cambridge, MA: National Bureau of Economic Research.
- Gehlbach, Scott, and Philip Keefer. 2011. "Investment without Democracy: Ruling-Party Institutionalization and Credible Commitment in Autocracies." *Journal of Comparative Economics* 39 (2): 123-139.
- Yanikkaya, Halit. 2003. "Trade Openness and Economic Growth: A Cross-Country Empirical Investigation." *Journal of Development Economics* 72 (1): 57-89

Tables and Figures

Table 1: Description of variables

name	varlab
country	Country
country_id	Country code
code	Country code
year	Year
rgdppc	Real GDP per capita (constant 2015 USD)
ln_gdp_pc	Log of real GDP per capita (constant 2015 USD)
growthrate	Real GDP per capita growth rate
democracy	Democracy index ranges from -10 to 10
stdDem	Standardized Democracy Index
gfcf	Gross fixed capital formation (% of GDP), proxy for investment
gfcf_missing	=1 if the origin value of gfcf is missing
tradeopen	Trade openness: (exports + imports) as % of GDP
tradeopen_missing	=1 if the origin value of tradeopen is missing
educ_missing	=1 if the origin value of educ is missing
educ	Average years of schooling from Barro-Lee dataset

Table 2: Summary Statistics

	Mean	SD	Min	Max	N
ln gdp pc	7.025	.922	5.135	10.927	2492
growthrate	1.684	6.232	-66.221	87.751	2438
stdDem	0	1	-1.453	1.782	2492
gfcf	16.063	13.227	-2.424	93.547	2492
gfcf missing	.27	.444	0.000	1	2492
tradeopen	59.727	54.316	0.000	437.327	2492
tradeopen missing	.128	.334	0.000	1	2492
educ	2.489	2.846	0.000	12.54	2492
educ missing	.416	.493	0.000	1	2492

Table 3: Detail information about Growth rate and stdDem

Variables	Obs	Mean	Std. Dev.	Min	Max	p1	p99	Skew.	Kurt.
growthrate	2438	1.684	6.232	-66.221	87.751	-15.745	16.348	-.497	39.155
stdDem	2492	0	1	-1.453	1.782	-1.453	1.782	.287	1.578

Figure 1: Growth Rate and Democracy

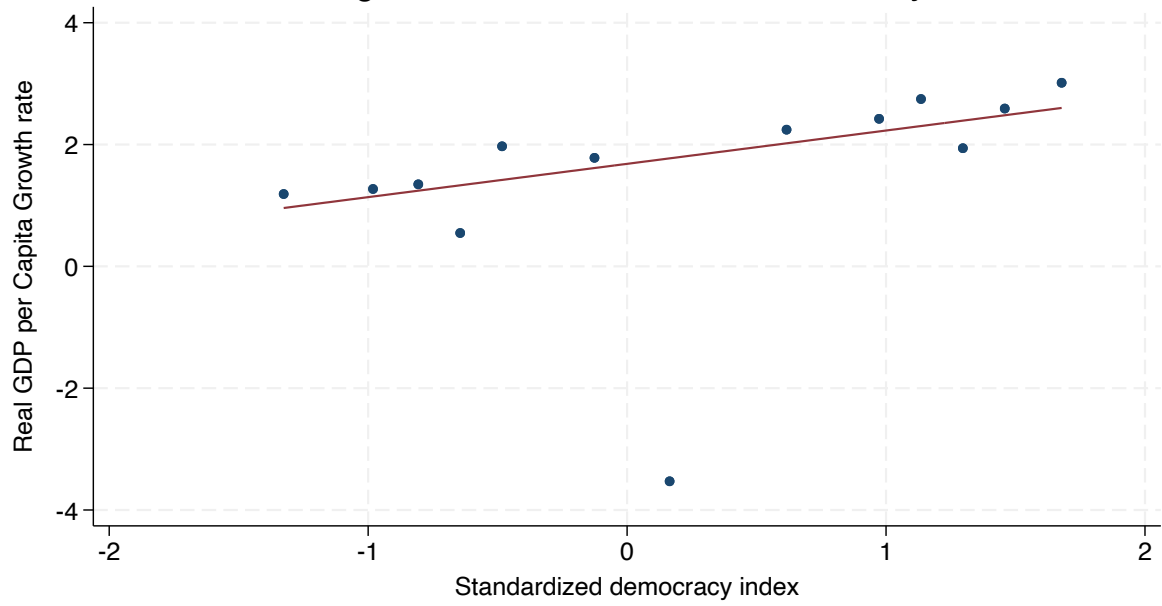


Figure 2: Nonlinear Fit: Growth rate and Democracy

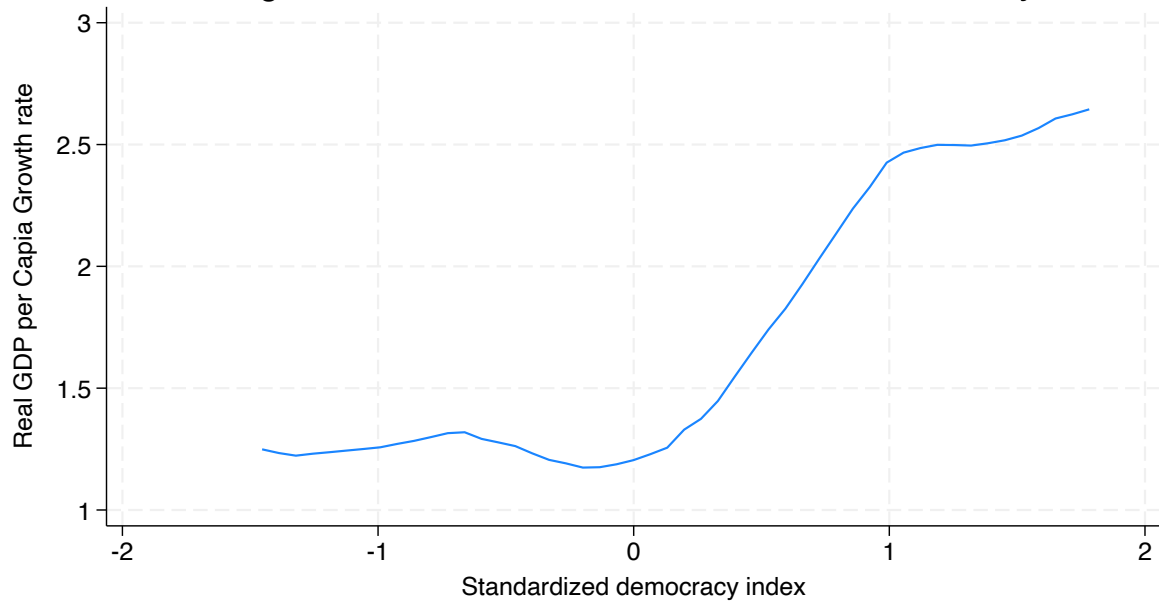


Table 4: Fisher-Type Panel Unit-Root Test for Log Real GDP per Capita

Test statistic	Statistic	p-value
Inverse chi-squared P	72.745	0.996
Inverse normal Z	5.286	1.000
Inverse logit L*	5.486	1.000
Modified inverse chi-squared Pm	-2.399	0.992

Notes: Notes: The test fails to reject the null hypothesis that all panels contain unit roots.

Table 5: Fisher-Type Panel Unit-Root Test for Real GDP Growth Rate

Test statistic	Statistic	p-value
Inverse chi-squared P	857.823	0.000
Inverse normal Z	-23.217	0.000
Inverse logit L*	-32.746	0.000
Modified inverse chi-squared Pm	51.019	0.000

Notes: The test rejects the null hypothesis that all panels contain unit roots.

Table 6: Growth Lag Checks

VARIABLES	(1) Lag 1	(2) Lag 1-2	(3) Lag 1-3	(4) Lag 1-4	(5) Lag 1-5	(6) Lag 1-6
stdDem	0.412** (0.173)	0.420** (0.162)	0.413*** (0.153)	0.458*** (0.158)	0.463*** (0.161)	0.468*** (0.174)
L.growthrate	0.130* (0.0756)	0.146** (0.0712)	0.135* (0.0714)	0.125* (0.0693)	0.125* (0.0698)	0.123 (0.0756)
L2.growthrate		0.0262 (0.0314)	0.0225 (0.0237)	0.0215 (0.0225)	0.0226 (0.0224)	0.0178 (0.0233)
L3.growthrate			0.0965*** (0.0341)	0.0911*** (0.0294)	0.0926*** (0.0301)	0.0894*** (0.0328)
L4.growthrate				0.0187 (0.0537)	0.0131 (0.0518)	0.00998 (0.0522)
L5.growthrate					0.00722 (0.0317)	0.0129 (0.0353)
L6.growthrate						-0.0339 (0.0332)
educ	0.129 (0.134)	0.133 (0.138)	0.160 (0.132)	0.151 (0.137)	0.110 (0.141)	0.136 (0.152)
gfcf	0.0965*** (0.0278)	0.0893*** (0.0295)	0.0762** (0.0301)	0.0732** (0.0330)	0.0642* (0.0346)	0.0660* (0.0349)
tradeopen	-0.0117 (0.00911)	-0.0119 (0.00902)	-0.0117 (0.00885)	-0.0124 (0.00906)	-0.0132 (0.00863)	-0.0134 (0.00916)
Constant	1.803 (1.458)	1.045 (1.257)	1.414 (1.257)	0.184 (1.379)	1.152 (1.214)	0.134 (1.395)
Observations	2,384	2,330	2,276	2,222	2,170	2,118
R-squared	0.089	0.094	0.099	0.096	0.094	0.092
Number of country_id	54	54	54	52	52	52
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Missing Indicators	Yes	Yes	Yes	Yes	Yes	Yes
SE clustered by country	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 7: Democracy and GDP Growth

VARIABLES	(1) No FE	(2) Country FE	(3) Country FE + Year FE
stdDem	0.344*** (0.108)	0.438*** (0.132)	0.413*** (0.153)
L.growthrate	0.187*** (0.0643)	0.143** (0.0675)	0.135* (0.0714)
L2.growthrate	0.0682*** (0.0200)	0.0313 (0.0216)	0.0225 (0.0237)
L3.growthrate	0.135*** (0.0277)	0.0897*** (0.0298)	0.0965*** (0.0341)
educ	0.0778 (0.0579)	0.0155 (0.0873)	0.160 (0.132)
gfcf	0.0534** (0.0244)	0.0619** (0.0286)	0.0762** (0.0301)
tradeopen	0.00434* (0.00223)	-0.00598 (0.00845)	-0.0117 (0.00885)
Constant	-0.913** (0.448)	-0.179 (0.969)	1.414 (1.257)
Observations	2,276	2,276	2,276
R-squared	0.119	0.050	0.099
Country FE	No	Yes	Yes
Year FE	No	No	Yes
Growth Lags	3	3	3
Controls	Yes	Yes	Yes
Missing Indicators	Yes	Yes	Yes
SE clustered by country	Yes	Yes	Yes
Number of country_id		54	54

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 8: Growth Lag Checks (Model 2)

VARIABLES	(1) Lag 1	(2) Lag 1-2	(3) Lag 1-3	(4) Lag 1-4	(5) Lag 1-5
stdDem	0.289 (0.257)	0.295 (0.248)	0.319 (0.239)	0.335 (0.234)	0.316 (0.238)
educ	0.0718 (0.162)	0.0725 (0.170)	0.112 (0.167)	0.0869 (0.169)	0.0329 (0.167)
c.stdDem#c.educ	0.0447 (0.0564)	0.0450 (0.0559)	0.0334 (0.0542)	0.0434 (0.0552)	0.0515 (0.0539)
L.growthrate	0.130* (0.0756)	0.145** (0.0712)	0.135* (0.0714)	0.125* (0.0693)	0.124* (0.0698)
L2.growthrate		0.0259 (0.0313)	0.0224 (0.0236)	0.0214 (0.0224)	0.0224 (0.0223)
L3.growthrate			0.0963*** (0.0340)	0.0908*** (0.0293)	0.0923*** (0.0300)
L4.growthrate				0.0185 (0.0537)	0.0129 (0.0518)
L5.growthrate					0.00710 (0.0317)
gfcf	0.0963*** (0.0278)	0.0890*** (0.0295)	0.0760** (0.0301)	0.0730** (0.0330)	0.0639* (0.0345)
tradeopen	-0.0119 (0.00918)	-0.0121 (0.00908)	-0.0119 (0.00889)	-0.0126 (0.00908)	-0.0134 (0.00864)
Constant	1.916 (1.492)	1.181 (1.311)	1.556 (1.345)	0.371 (1.456)	1.380 (1.253)
Observations	2,384	2,330	2,276	2,222	2,170
R-squared	0.089	0.094	0.099	0.096	0.094
Number of country_id	54	54	54	52	52
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Missing Indicators	Yes	Yes	Yes	Yes	Yes
SE clustered by country	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 9: Democracy and GDP Growth (Model 2)

VARIABLES	(1) No FE	(2) Country FE	(3) Country FE + Year FE
stdDem	0.263 (0.162)	0.297 (0.180)	0.319 (0.239)
educ	0.0516 (0.0706)	-0.0444 (0.0980)	0.112 (0.167)
c.stdDem#c.educ	0.0326 (0.0433)	0.0592 (0.0441)	0.0334 (0.0542)
L.growthrate	0.187*** (0.0643)	0.143** (0.0676)	0.135* (0.0714)
L2.growthrate	0.0680*** (0.0199)	0.0309 (0.0217)	0.0224 (0.0236)
L3.growthrate	0.135*** (0.0277)	0.0891*** (0.0300)	0.0963*** (0.0340)
gfcf	0.0526** (0.0247)	0.0621** (0.0286)	0.0760** (0.0301)
tradeopen	0.00438* (0.00221)	-0.00619 (0.00838)	-0.0119 (0.00889)
Constant	-0.832* (0.481)	0.0979 (0.980)	1.556 (1.345)
Observations	2,276	2,276	2,276
R-squared	0.120	0.050	0.099
Country FE	No	Yes	Yes
Year FE	No	No	Yes
Growth Lags	3	3	3
Controls	Yes	Yes	Yes
Missing Indicators	Yes	Yes	Yes
SE clustered by country	Yes	Yes	Yes
Number of country_id		54	54

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 10: Baseline regression for model 1 and model 2

VARIABLES	(1) Model 1	(2) Model 2
stdDem	0.413*** (0.153)	0.319 (0.239)
educ	0.160 (0.132)	0.112 (0.167)
c.stdDem#c.educ		0.0334 (0.0542)
L.growthrate	0.135* (0.0714)	0.135* (0.0714)
L2.growthrate	0.0225 (0.0237)	0.0224 (0.0236)
L3.growthrate	0.0965*** (0.0341)	0.0963*** (0.0340)
gfcf	0.0762** (0.0301)	0.0760** (0.0301)
tradeopen	-0.0117 (0.00885)	-0.0119 (0.00889)
Constant	1.414 (1.257)	1.556 (1.345)
Observations	2,276	2,276
R-squared	0.099	0.099
Number of country_id	54	54
Country FE	Yes	Yes
Year FE	Yes	Yes
Growth Lags	3	3
Controls	Yes	Yes
Missing Indicators	Yes	Yes
SE clustered by country	Yes	Yes

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 11: Growth Rate and stdDem Lag Checks

VARIABLES	(1) growthrate	(2) growthrate	(3) growthrate	(4) growthrate
stdDem	0.370 (0.403)	0.378 (0.408)	0.364 (0.402)	0.380 (0.378)
L.stdDem	0.0434 (0.555)	-0.00383 (0.559)	0.0151 (0.543)	0.0126 (0.512)
L2.stdDem	0.0642 (0.468)	0.0459 (0.455)	0.0305 (0.450)	0.0141 (0.453)
L3.stdDem	-0.598 (0.509)	-0.512 (0.489)	-0.508 (0.493)	-0.565 (0.505)
L4.stdDem	0.442 (0.432)	0.384 (0.425)	0.393 (0.427)	0.652* (0.353)
L5.stdDem	0.271 (0.318)	0.310 (0.320)	0.309 (0.320)	
L.growthrate	0.124* (0.0700)	0.125* (0.0694)	0.126* (0.0728)	0.135* (0.0714)
L2.growthrate	0.0225 (0.0223)	0.0214 (0.0224)	0.0216 (0.0233)	0.0226 (0.0235)
L3.growthrate	0.0930*** (0.0301)	0.0912*** (0.0293)	0.0936*** (0.0333)	0.0966*** (0.0340)
L4.growthrate	0.0123 (0.0524)	0.0178 (0.0543)		
L5.growthrate	0.00585 (0.0322)			
educ	0.118 (0.140)	0.160 (0.137)	0.160 (0.139)	0.166 (0.131)
gfcf	0.0655* (0.0345)	0.0745** (0.0328)	0.0761** (0.0298)	0.0770** (0.0297)
tradeopen	-0.0138 (0.00863)	-0.0131 (0.00903)	-0.0132 (0.00926)	-0.0121 (0.00884)
Constant	1.086 (1.250)	0.157 (1.387)	0.182 (1.403)	1.426 (1.266)
Observations	2,170	2,222	2,222	2,276
R-squared	0.096	0.097	0.097	0.100
Number of country_id	52	52	52	54
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Missing Indicators	Yes	Yes	Yes	Yes
SE clustered by country	Yes	Yes	Yes	Yes

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 11: Continue

VARIABLES	(1) growthrate	(2) growthrate	(3) growthrate	(4) growthrate
stdDem	0.396 (0.379)	0.398 (0.378)	0.398 (0.377)	0.413*** (0.153)
L.stdDem	0.0310 (0.514)	0.0290 (0.513)	0.0176 (0.362)	
L2.stdDem	-0.0356 (0.452)	-0.0127 (0.330)		
L3.stdDem	0.0253 (0.308)			
L.growthrate	0.135* (0.0715)	0.135* (0.0715)	0.135* (0.0714)	0.135* (0.0714)
L2.growthrate	0.0225 (0.0236)	0.0225 (0.0236)	0.0225 (0.0237)	0.0225 (0.0237)
L3.growthrate	0.0965*** (0.0340)	0.0965*** (0.0341)	0.0965*** (0.0341)	0.0965*** (0.0341)
educ	0.161 (0.132)	0.160 (0.132)	0.160 (0.132)	0.160 (0.132)
gfcf	0.0762** (0.0299)	0.0762** (0.0300)	0.0762** (0.0301)	0.0762** (0.0301)
tradeopen	-0.0118 (0.00886)	-0.0118 (0.00889)	-0.0118 (0.00886)	-0.0117 (0.00885)
Constant	1.414 (1.262)	1.415 (1.259)	1.414 (1.257)	1.414 (1.257)
Observations	2,276	2,276	2,276	2,276
R-squared	0.099	0.099	0.099	0.099
Number of country_id	54	54	54	54
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Missing Indicators	Yes	Yes	Yes	Yes
SE clustered by country	Yes	Yes	Yes	Yes

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1